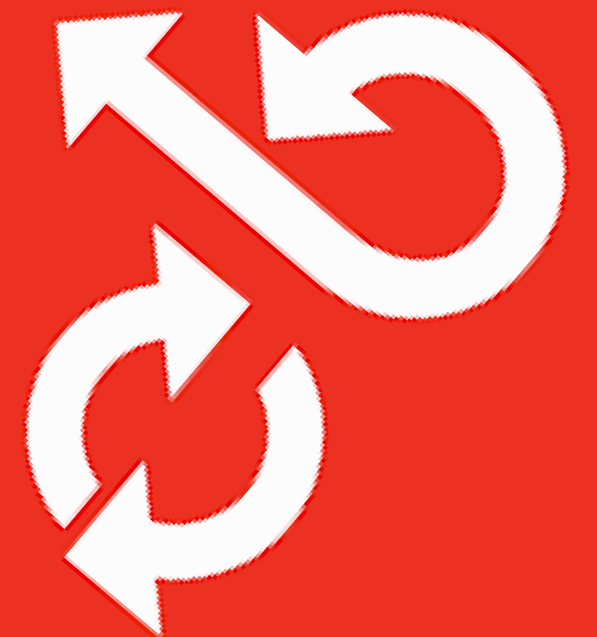


Patterns, Trends and AI Visuals: The Capital Bikeshare History

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Date: 20.07.2026



Executive summary

The whole capstone, in one breath

We combined more than 3 years of Capital Bikeshare trip data with NOAA weather records and station-level detail into a single Power BI model.

After cleaning millions of rows of ridership data and reconciling two different weather file formats.

The analysis surfaced a clear driver of demand merged with Weather impact. Then Found a striking anomaly of over 1.6 million same-station round trips and produced a validated 2026 ridership forecast that we tested directly against actual results.

The result is a working dashboard that shows where riders start, where they go and how season shape that behavior.

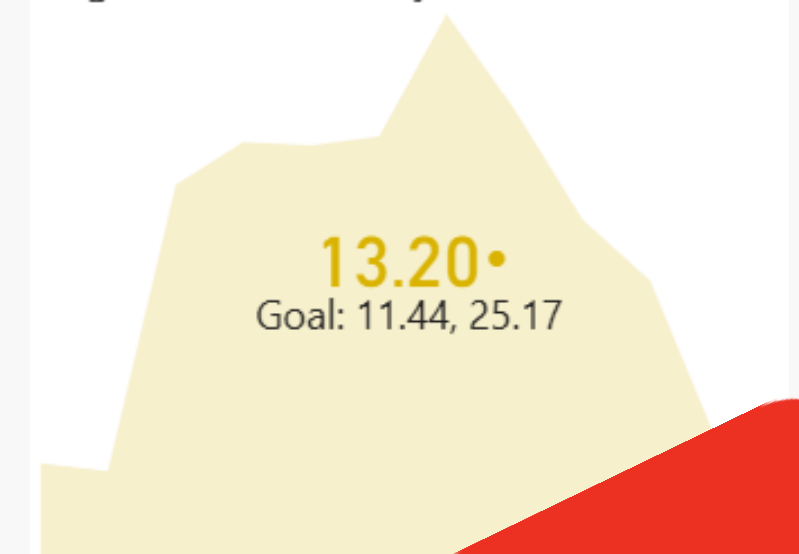
20M total rides analyzed across a multi-year, multi-source model.

KPI Numbers

Quantity of rides by Month



Avg Duration (min) by Month



Context

What Capital Bikeshare is, and what this analysis covers

Capital Bikeshare is the public bikeshare system serving Washington D.C. and the surrounding region. Riders check out a classic, docked or electric bike at one station and return it at another with usage tracked at the trip level for every ride.

Period of this analysis: January 2019 – June 2026

Trip-level ridership joined with daily weather, spanning normal seasons, a global disruption, and recovery.

- **Two rider types**

Members (subscribers) and casual (single-ride)

- **Three bike types**

Classic, docked and electric bikes

- **Dock-based system**

Pick up at one station, return at another

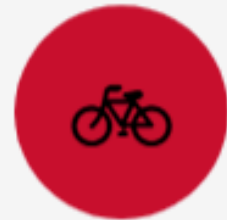
- **Weather-sensitive**

Ridership tracks temperature and precipitation



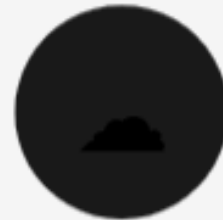
Data Sources

Three sources, one model



Ridership data

Capital Bikeshare trip-level records, 2023–2026: 4.47M rows (2023), 6.11M (2024), 6.66M (2025), 2.85M (2026 partial).



Weather data

Daily temperature and precipitation from NOAA (ncei.noaa.gov), merged from two differently-formatted file sets into one clean time series.



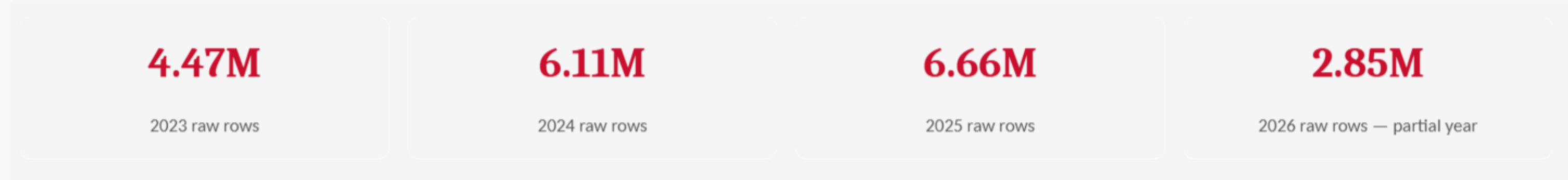
Station info

Station-level detail grouped by hour with member/casual counts, joined to ridership and weather to complete the model.



Data quality

What cleaning was needed and why it mattered



- Combining and grouping all years by hour in Power BI took roughly 12.5 hours of calculation on its own.
- Under 1% of rows had missing station data - 53 of 30,696 missing end stations, 44 of 30,696 missing start stations.
- Weather data spanned two different file formats (Jan 2023-Aug 2025 and Aug 2025-Jun 2026) and was combined from a folder of CSVs into one clean Datetime, Temperature and Precipitation table before merging with ride data.
- Station-level grouping by hour with member/casual counts (bike type unavailable) was too slow inside Power BI - 10 hours to calculate plus 5 hours to save. Fixed by grouping outside Power BI and importing the result as "Station Info."



The Anomaly

A stretch of history that looks off

1,606,470

rides start and end at the same station

Most visible on the Columbus Circle / Union Station route, where the start-to-end route is Columbus Circle / Union Station itself.

Reads like an anomaly:

On the surface, over 1.6 million identical-station round trips looks unusual. Far too large a share to dismiss as riders simply changing their minds.

Confirming the real cause: docking errors, loop routes, maintenance checks.

It needs a deeper join than Power BI can comfortably do here. That analysis belongs in a proper database, not this report.



Decomposition Tree

The clearest breakdown path

If drilled member rides down to their single highest-temperature day is August 9, 2024 at 78°F.

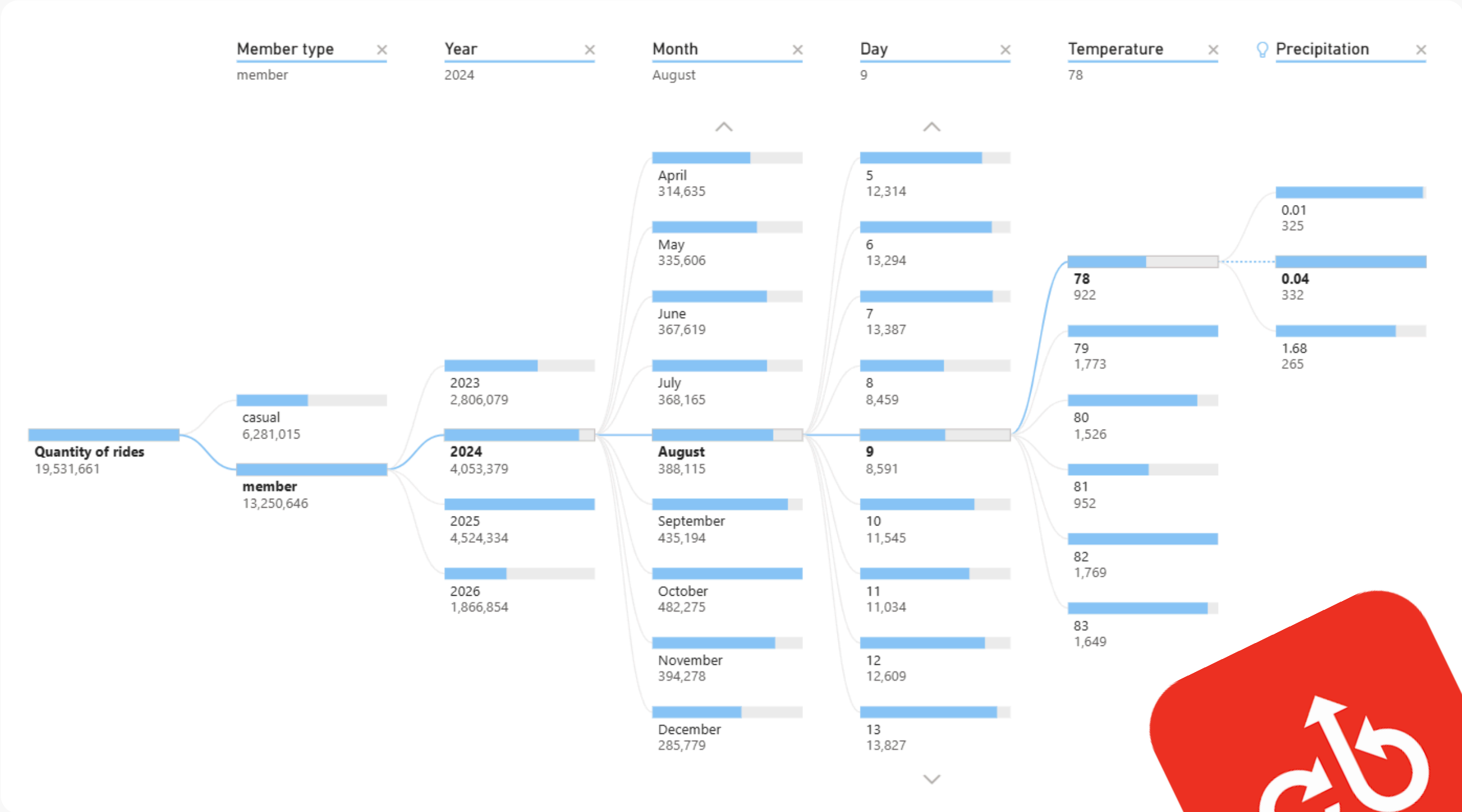
To isolate how much of the swing in ride count temperature alone explains before precipitation is layered on.

Path:

Member →

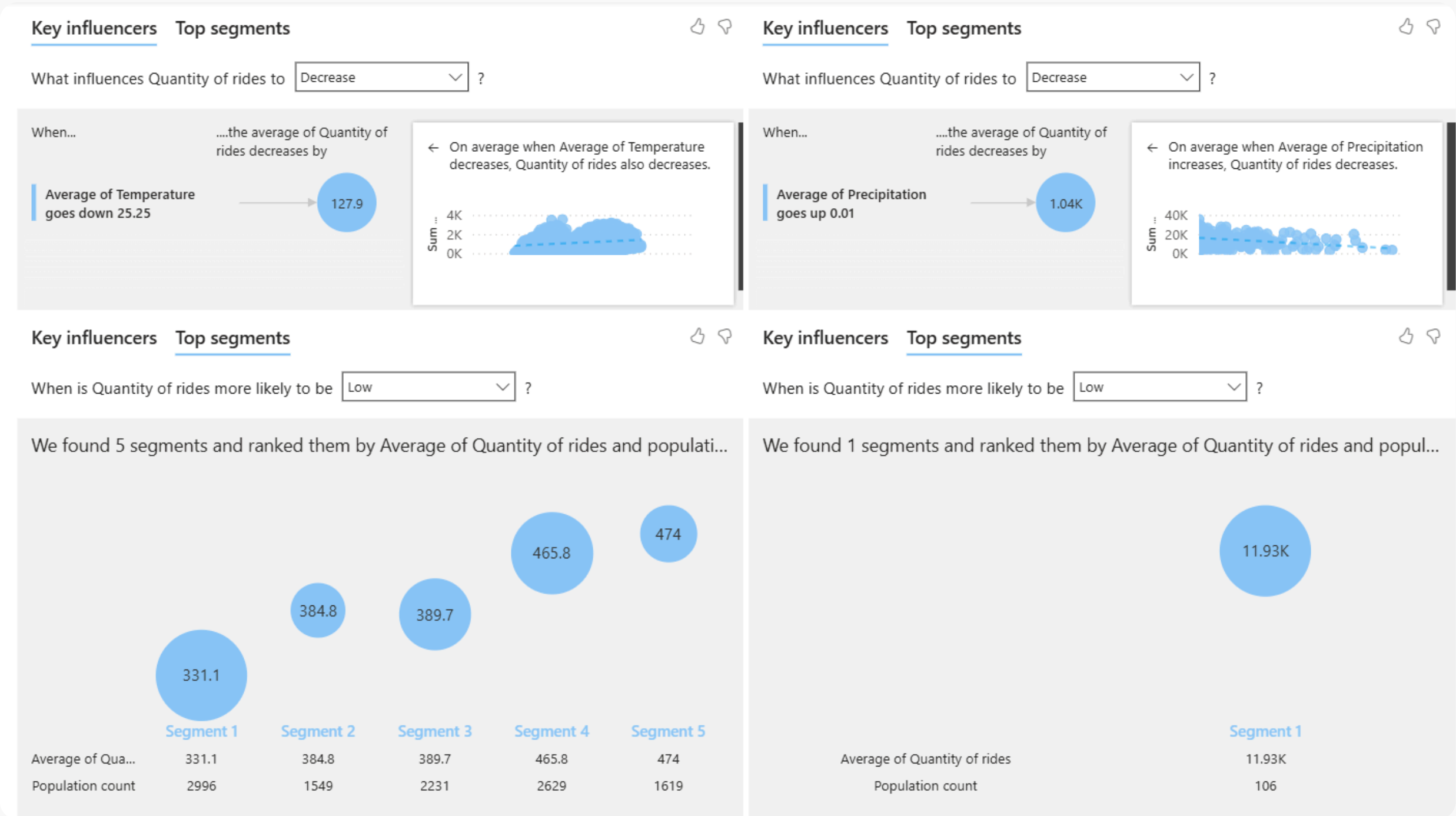
2024 → August → Day 9 →

78°F (peak) → Precipitation



Key Influencers

Weather is the strongest driver



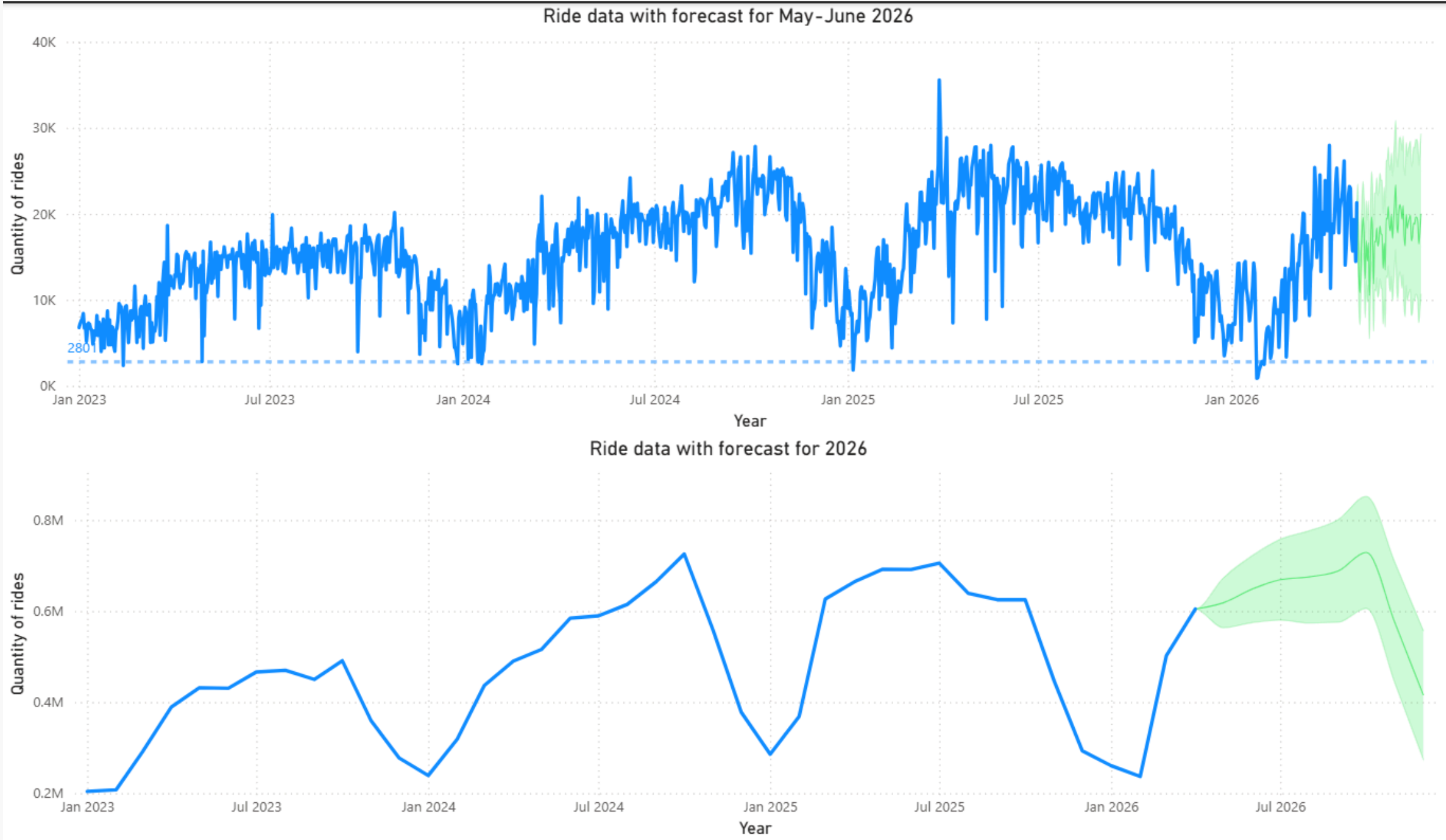
Temperature: When average temperature drops 25.25°, average ride volume falls by roughly 128 rides Its a large, consistent effect.

Precipitation: Rising precipitation also decreases rides, but the scatter is far less clean. We treat this signal with lower confidence than temperature.



Forecast, Before

The original forecast



Why it didn't hold up:

The first pass was built only from the last two months of daily data, filtered out on its own and projected forward.

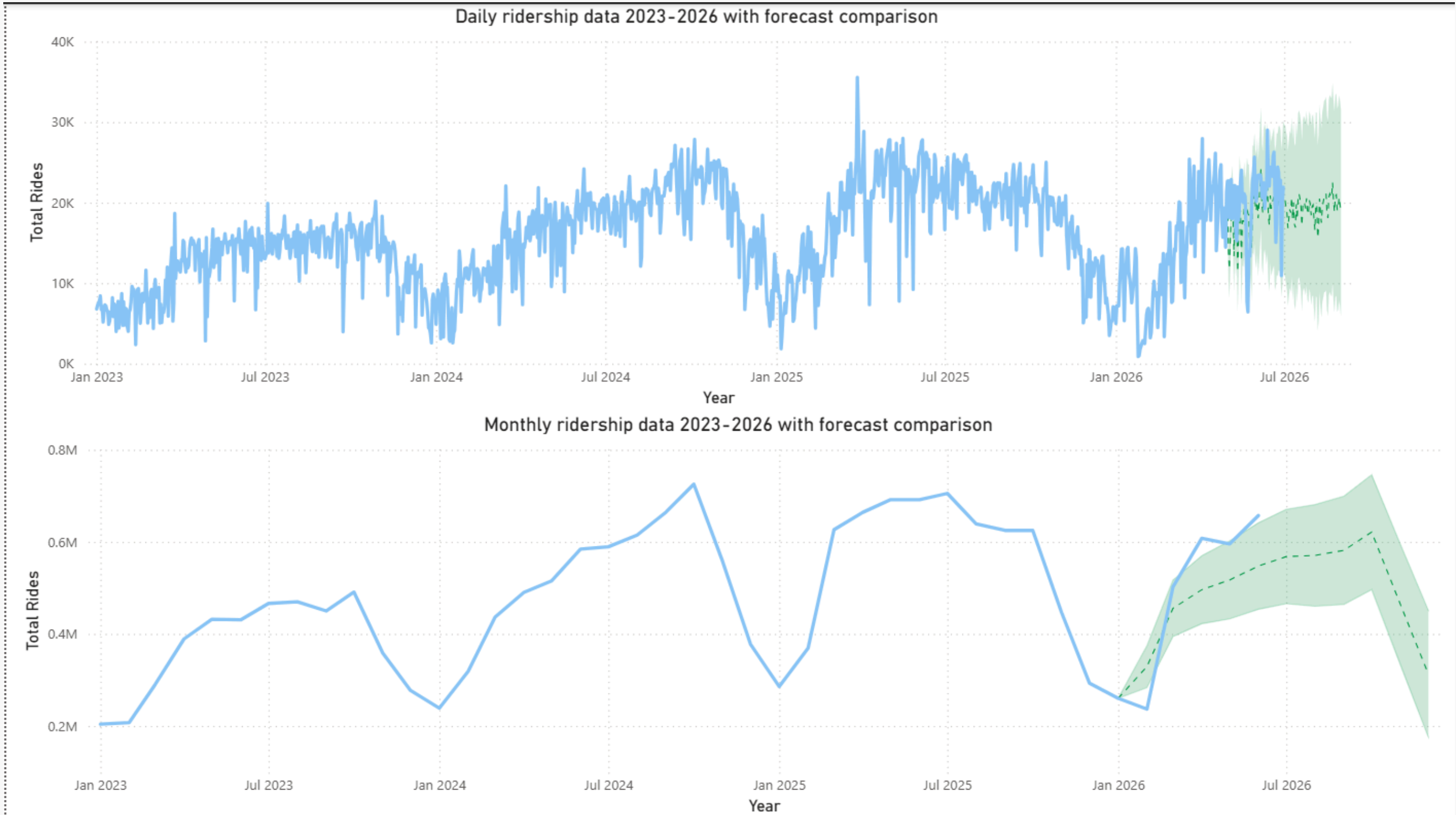
With no earlier context in the training window, the monthly projection swings up to nearly 75% its an optimistic trend with a wide, unvalidated confidence band.

Nothing here was checked against real, observed results yet.



Forecast, After

The revised forecast



I rebuilt the comparison to hold out actual daily data for the last two months and actual monthly data across all of 2026 and plotted both against the forecast.

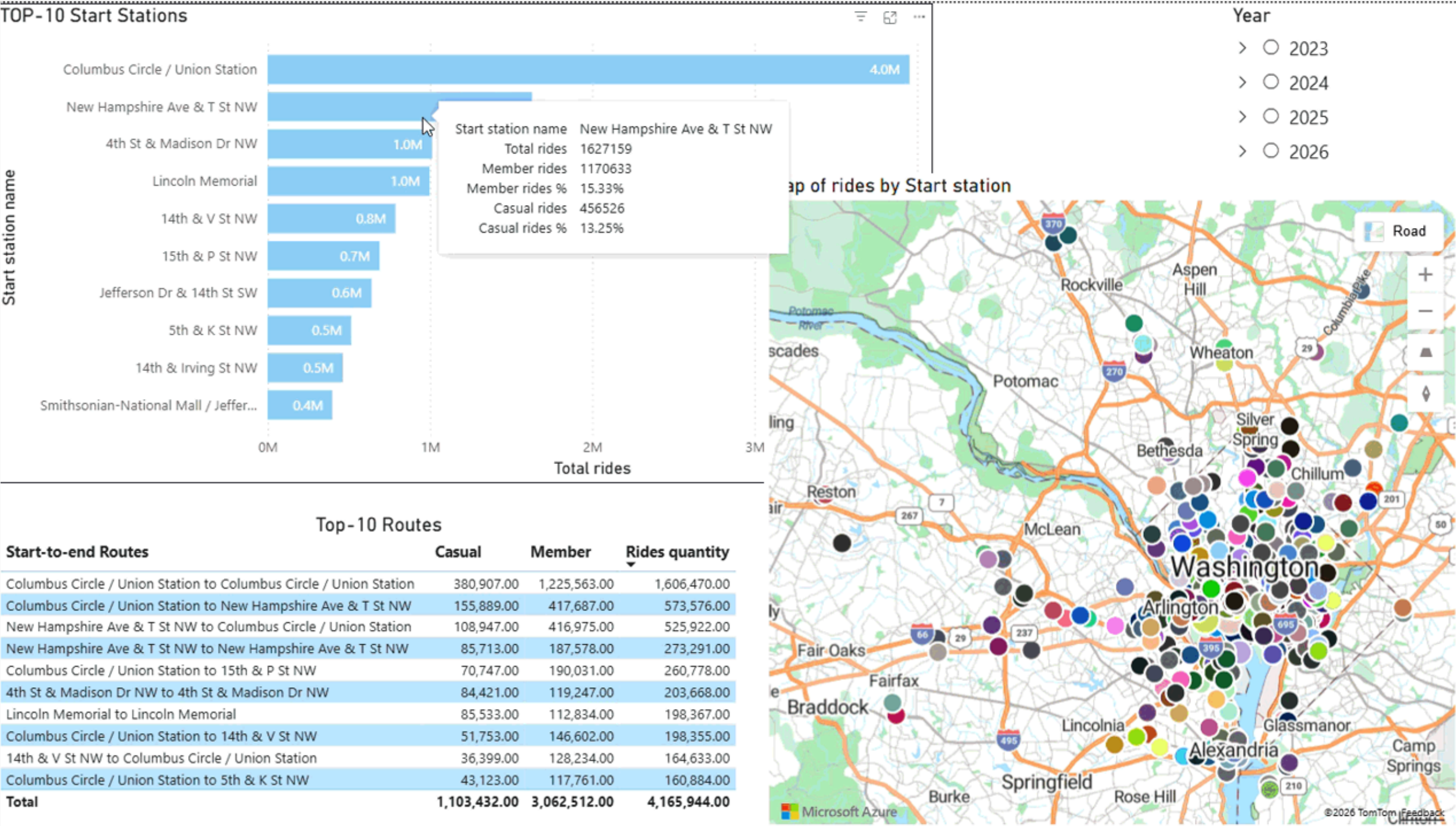
That gave us the chance to check the forecast against what actually happened - the dashed actual line runs below the original band, showing the first forecast overshoot.

Validated against reality, not just projected.



Station and route analysis

Busiest stations and routes



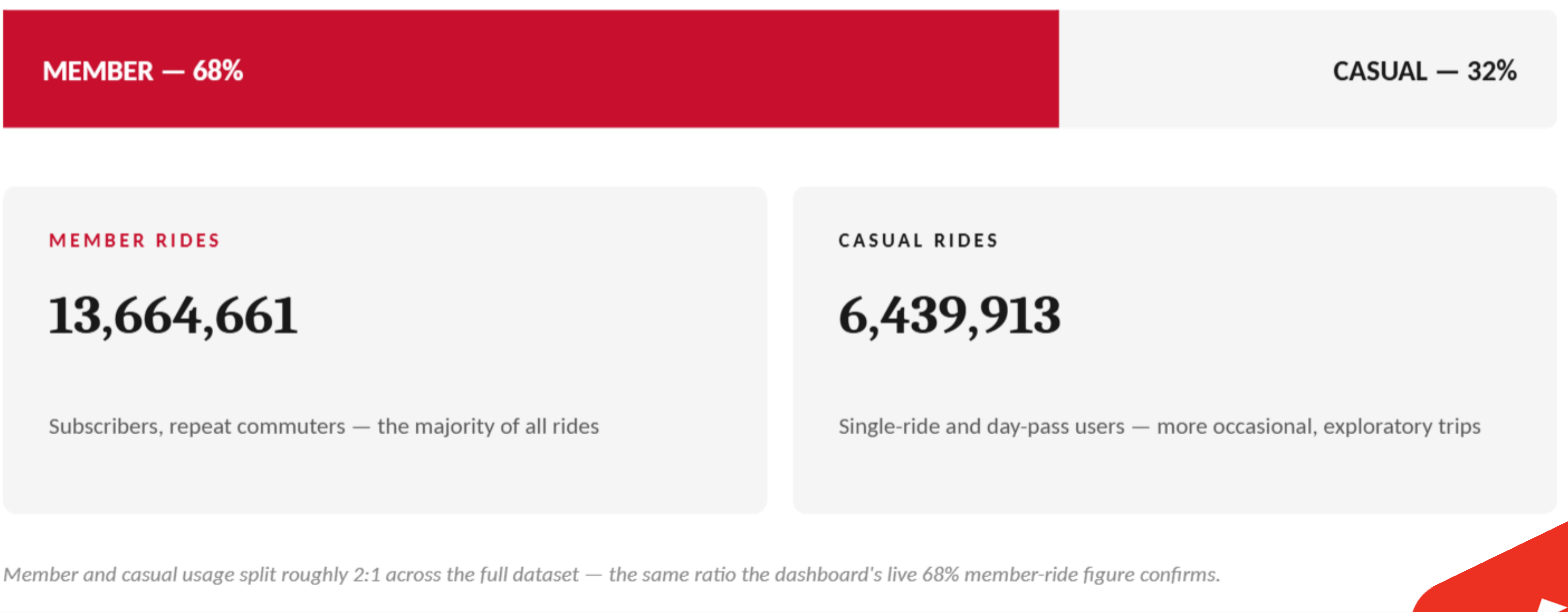
Side-by-side visuals :

- Bar char Top-10 busiest start stations
- Map with all stations info
- Table Top-10 busiest routes ride counts.



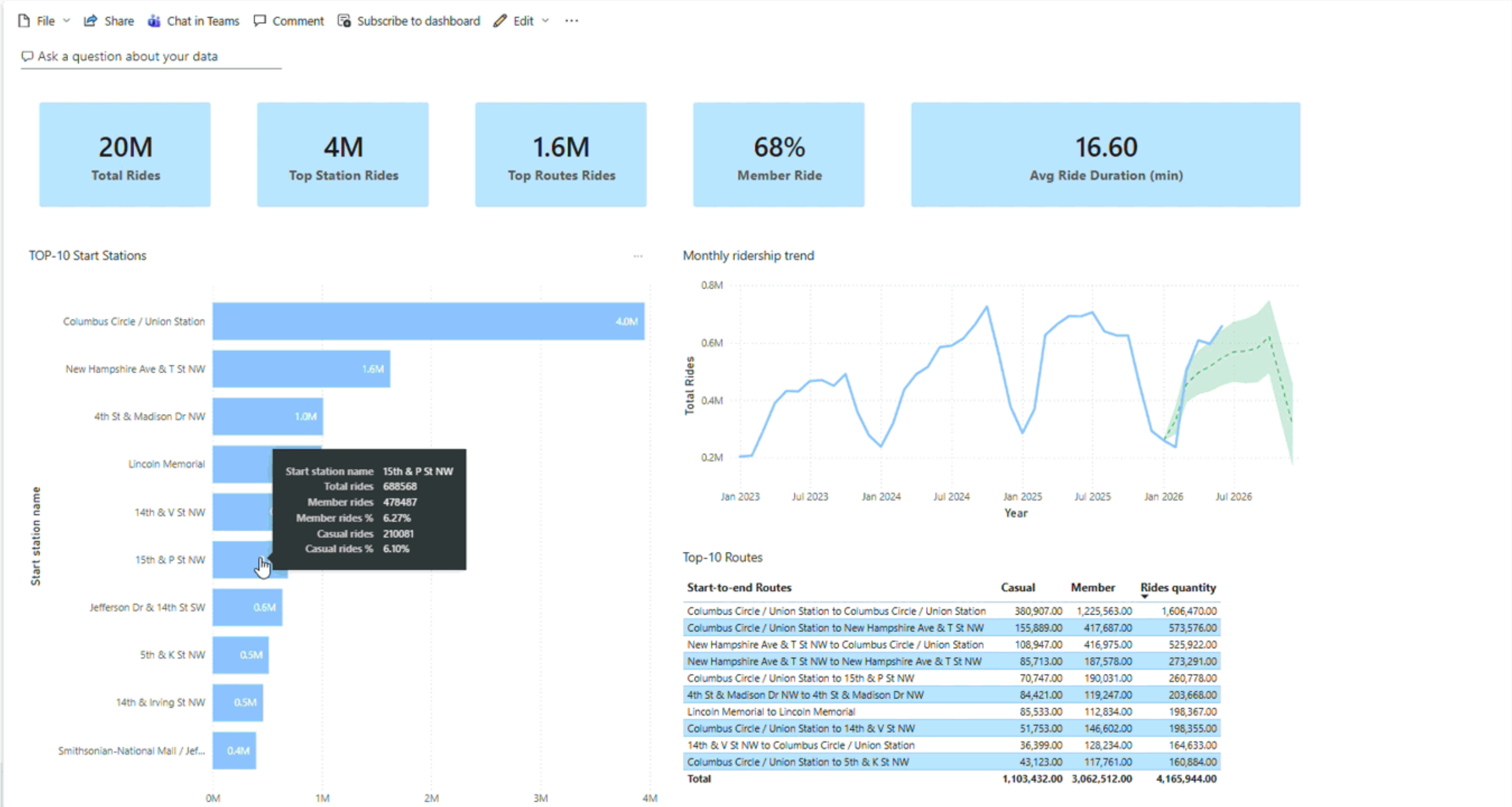
Member vs. casual mix

Rider Segments



Dashboard walkthrough

The finished dashboard, live



Power BI dashboard :

- KPI cards up top
- Bar char Top-10 busiest start stations
- Monthly ridership trend line with forecast overlay
- Table Top-10 busiest routes ride counts



Lessons Learned

What we'd do differently

Power BI is a reporting tool, not a data engineering platform:

Grouping raw ridership data, well over a million rows, directly inside Power BI / Power Query was the wrong call. It's what drove the 12+ hour calculation times and for the station-level grouping a 10-hour compute plus a further 5 hours just to save the result.

With more time, that heavy grouping and transformation would move to a proper data platform. Something like Snowflake or Databricks to built for large-scale processing, leaving Power BI to do what it does best: reporting and visualization on top of clean, pre-aggregated data.



Close

Ready for the next build

The clear next step is starting the ETL in a real database instead of loading raw trip data straight into Power Query.

That gives us the power to rebuild the route analysis faster and to properly resolve the same-station outliers this report surfaced, rather than working around Power BI's limits.

- Move heavy grouping to a database-backed ETL
- Re-run route analysis on the cleaner pipeline.
- Confirm the cause behind same-station outliers



Thank you!

